

2.5 Continuity

Recall Def: $f(x)$ is continuous at $x=c$ if

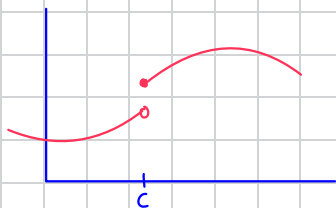
- $\lim_{x \rightarrow c} f(x), f(c)$ exist
- $\lim_{x \rightarrow c} f(x) = f(c)$

Understand maths geometrically and physically

otherwise, we say $f(x)$ is discontinuous at $x=c$

Def If $\lim_{x \rightarrow c^+} f(x) = f(c)$, then we say $f(x)$ is right-continuous at $x=c$
 If $\lim_{x \rightarrow c^-} f(x) = f(c)$, then we say $f(x)$ is left-continuous at $x=c$

Example



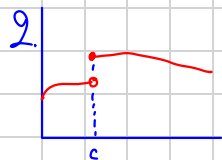
$f(x)$ is right continuous



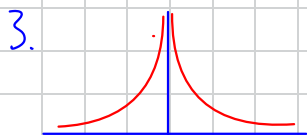
- All polynomials are continuous at every $x=c$.
- All trig functions $\sin x, \cos x, \tan x, \cot x, \csc x, \sec x$ are continuous at every c in their domains.

Types of discontinuity

1. Either $f(c)$ or $\lim_{x \rightarrow c} f(x)$ exist but $\neq f(c) \rightarrow$ Removable discontinuity at $x=c$
 $f_{\text{new}}(c) = \begin{cases} f(x) & ; x \neq c \\ \lim_{x \rightarrow c} f(x) & ; x = c \end{cases}$

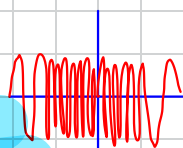


$\lim_{x \rightarrow c^+} f(x) \neq \lim_{x \rightarrow c^-} f(x)$
 $\rightarrow$ We say $f(x)$ has jump discontinuity



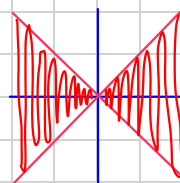
e.g. $f(x) = \frac{1}{x^2}$
 $f(x)$ blows up as $x \rightarrow 0$
 Then we say $f(x)$ is infinite discontinuity at $x=c$

4. e.g. $f(x) = \sin(\frac{1}{x})$
 $f(x)$ oscillates infinitely with amplitude $\geq$ fixed positive number
 $\rightarrow f$ has oscillating discontinuity



$f(x) = x \sin(\frac{1}{x})$
 $\lim_{x \rightarrow 0} f(x) = 0, f(0) \text{ DNE}$
 $\therefore f_{\text{new}}(x) = \begin{cases} x \sin(\frac{1}{x}) & ; x \neq 0 \\ 0 & ; x = 0 \end{cases}$

Removable disc. at $x=0$



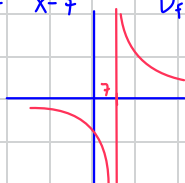
Def we say f is continuous on $[a, b]$ iff

- f is continuous in every $c \in (a, b)$
- f is left-continuous at $x = b$ ($\lim_{x \rightarrow b^-} f(x) = f(b)$)
- f is right-continuous at $x = a$ ($\lim_{x \rightarrow a^+} f(x) = f(a)$)



Otherwise, we say $f(x)$ is discontinuous at $[a, b]$

e.g. $f(x) = \frac{1}{x-7}$ $D_f = \{x \in \mathbb{R} \mid x \neq 7\}$



→ continuous function on every c in their domains

Building complicated continuous function from simple ones

Theorem 1 Suppose $f(x), g(x)$ are continuous at $x=c$ then

1) $f(x) + g(x)$ is continuous at $x=c$

→ $\lim_{x \rightarrow c} f(x) + g(x) \stackrel{\text{Sum Rule}}{=} \lim_{x \rightarrow c} f(x) + \lim_{x \rightarrow c} g(x) = f(c) + g(c) = f(x) + g(x)|_{x=c}$

2) $f(x)g(x)$ is continuous at $x=c$

3) $\frac{f(x)}{g(x)}$ is continuous at $x=c$ if $g(c) \neq 0$

↳ in particular, $\frac{1}{g(x)}$ is continuous

4) $(f(x))^n$ is continuous at $x=c$ ($n \in \mathbb{N}$)

5) $\sqrt[n]{f(x)}$ is continuous at $x=c$ (provided $\sqrt[n]{f(c)}$ defined) → not imaginary

Theorem 2 Suppose $f(x)$ is continuous at $x=c$ and $g(x)$ is continuous at $x=f(c)$ then composite $g \circ f(x) = g(f(x))$ is also continuous at $x=c$

e.g. $f(x) = \sin(\cos x)$ continuous at every x

Proof since f is continuous at c , we have $f(x) \rightarrow f(c)$ as $x \rightarrow c$;

Since g is continuous at $f(c)$, we have

$g(w) \rightarrow g(f(c))$ as $w \rightarrow f(c)$ ($w = f(x)$)

Thm 1 + Thm 2 → All elementary functions (Polynomials, Trigonometry, Rational functions, root, log, expo) and their combinations in the fashion of Thm 1 are continuous on their domain

2.5 (34) $\lim_{x \rightarrow 0} \tan\left(\frac{\pi}{4} \cos(\sin x^{1/3})\right) = \tan\left(\frac{\pi}{4} \cos(\sin 0)\right) = \tan\left(\frac{\pi}{4} \cos 0\right) = \tan\left(\frac{\pi}{4} \cdot 1\right) = 1$

Intermediate value Theorem

→ Suppose $f(x)$ is continuous on $[a, b]$

If d is between $f(a)$ and $f(b)$ $[f(a) \leq d \leq f(b) \text{ or } f(b) \leq d \leq f(a)]$

then $\exists c \in [a, b]$ such that $f(c) = d$ $\rightarrow f(a)f(b) < 0$

Remark If $f(a)$ and $f(b)$ have opposite signs, then $\exists c \in (a, b)$

(f continuous) such that $f(c) = 0$

i.e. $f(x) = 0$ has a solution between a and b

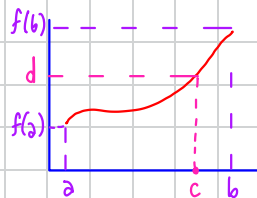
9.5 (54) $f(x) = \cos x - x$

$$f(0) = 1 - 0 = 1$$

$$f(\frac{\pi}{2}) = 0 - \frac{\pi}{2} = -\frac{\pi}{2}$$

} By IVT, $\exists c \in [0, \frac{\pi}{2})$ such that $f(c) = 0$ or $\cos c = c$

i.e. $\cos x = x$ has a solution



Counter example



No such c that $f(c) = d$

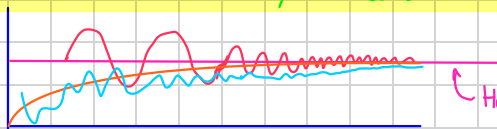
2.6 Limits involving infinity & Asymptotes of graph

Notation ∞ (infinity) means that a variable (either independent or dependent) is getting unstoppably large

Def We say a number L is the limit of $f(x)$ as $x \rightarrow \infty$ if

• $f(x)$ is defined at least on $[a, \infty)$ $a \in \mathbb{R}$

• $f(x)$ is close to L , as close as we wish, whenever x is large enough.



Horizontal asymptote of x

Def We say $\lim_{x \rightarrow c} f(x)$ is positive infinity if $f(x)$ is arbitrarily large, as large as we wish, when x is close (not equal) to c enough.

Written $\lim_{x \rightarrow c} f(x) = \pm \infty$

In this case, we say $x = c$ is vertical asymptote of x .



2.6(31): $\lim_{x \rightarrow \infty} \frac{2x^{5/3} - x^{1/3} + 7}{x^{8/5} + 3x + \sqrt{x}}$ secretly $= \lim_{x \rightarrow \infty} \frac{2x^{5/3}}{x^{8/5}} = \lim_{x \rightarrow \infty} 2x^{1/5} = \infty$

$$= \lim_{x \rightarrow \infty} \frac{x^{5/3} [2 - x^{-4/3} + 7x^{-5/3}]}{x^{8/5} [1 - x^{-3/5} + x^{-11/10}]}$$

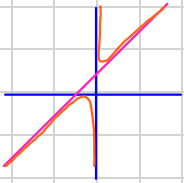
$$\lim_{x \rightarrow \infty} x^{-k} = 0 \quad (k \in \mathbb{R}^+)$$

$$= \lim_{x \rightarrow \infty} x^{1/5} \cdot \frac{2}{1} = \infty \cdot 2 = \infty$$

Oblique asymptotes

$$y = \frac{x^2+1}{x-1} = f(x)$$

$$x+1 + \frac{2}{x-1}$$



$$y = x+1$$

$$\lim_{x \rightarrow 1^+} f(x) = \frac{2}{0^+} = +\infty$$

$$\lim_{x \rightarrow 1^-} f(x) = \frac{2}{0^-} = -\infty$$

→ $x=1$ Vertical asymptote

Behavior of $f(x)$ as $x \rightarrow \infty$?

$$f(x) = \frac{x^2+1}{x-1} = x+1 + \frac{2}{x-1} \rightarrow 0$$

$f(x) \approx x+1$ when x tremendous

Oblique asymptote (tilted)